

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3) Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2) Question Count: 5

Total Marks: 25

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Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Introduction

A media company migrating its video streaming platform to the cloud must ensure high performance, scalability, and minimal buffering for millions of users.

Cloud Architecture

- Use cloud storage for videos.
- Deploy application servers behind a **Load Balancer**.
- Use **Content Delivery Network (CDN)** to deliver videos from edge locations.
- Store frequently accessed content in **cache** to reduce server load.

Load Balancing and Auto-Scaling

- Load balancer distributes user requests evenly across multiple servers.
- Auto-scaling automatically adds servers during peak traffic and removes them when demand decreases.
- This ensures uninterrupted streaming and prevents server overload.

Caching Techniques

- Cache popular videos at CDN edge servers.
- Users receive content from the nearest server, reducing latency and buffering.

Cost Optimization

- Auto-scaling avoids paying for unused servers.
- Use object storage for videos.
- Use reserved instances for regular workloads and spot instances for temporary processing.
- CDN reduces bandwidth costs.

Conclusion

A cloud architecture with CDN, caching, load balancing, and auto-scaling provides smooth streaming, high availability, and cost-efficient operation.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Introduction

A fintech startup deploys applications across multiple cloud providers using containers. The challenge is maintaining consistency and managing deployments efficiently.

Proposed Solution

- Package applications using **Docker containers**.
- Use **Kubernetes** to manage container deployment, scaling, and recovery across clouds.

Service Discovery

- Kubernetes automatically identifies and connects services.
- Applications communicate using service names instead of IP addresses.

Scaling

- Kubernetes automatically increases or decreases container replicas based on CPU or memory usage.
- Load balancing distributes requests among containers.

Benefits of Multi-Cloud

- High availability
- Disaster recovery
- Reduced vendor lock-in
- Better performance by selecting nearby cloud regions

Risks

- Complex management
- Increased networking costs
- Security policy differences
- Monitoring across multiple clouds

Conclusion

Using Docker and Kubernetes provides consistent deployment, automatic scaling, and high availability while reducing dependence on a single cloud provider.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Introduction

An e-learning platform accidentally loses important cloud data. A backup and disaster recovery strategy is needed.

Proposed Solution

- Schedule automatic cloud backups.
- Enable **versioning** to recover deleted or modified files.
- Replicate data to another cloud region.
- Store backups securely using cloud object storage.

RTO and RPO

- **RTO (Recovery Time Objective):** Maximum acceptable downtime. Example: Restore services within **1 hour**.
- **RPO (Recovery Point Objective):** Maximum acceptable data loss. Example: Restore data up to the last **15 minutes**.

Business Continuity

- Use multiple cloud regions.
- Perform regular backup testing.
- Monitor backup status.
- Automate recovery procedures.

Example

If course materials are accidentally deleted, versioning restores previous copies, while replicated backups ensure data remains available even if one region fails.

Conclusion

Cloud-native backup, replication, and versioning provide fast recovery, minimal data loss, and uninterrupted learning services.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Introduction

A government agency uses a hybrid cloud where sensitive applications remain on-premises, while non-sensitive applications run in the public cloud.

Hybrid Architecture

- Sensitive databases remain in the private data center.
- Public cloud hosts websites and public services.
- Connect both environments using secure VPN or dedicated private links.

Identity Management

- Use centralized Identity and Access Management (IAM).
- Implement Role-Based Access Control (RBAC).
- Enable Multi-Factor Authentication (MFA).

Encryption

- Encrypt stored data.
- Encrypt data transferred between on-premises and cloud using SSL/TLS.

Challenges

- Complex management
- Maintaining consistent security policies
- Network latency
- Compliance with government regulations

Conclusion

A hybrid cloud provides flexibility while ensuring sensitive government data remains secure through encryption, IAM, and secure communication.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Introduction

A SaaS provider must ensure its application remains available even if one cloud region fails.

Multi-Region Deployment

- Deploy application servers in multiple cloud regions.
- Use a global load balancer to direct users to the nearest healthy region.

Data Replication

- Replicate databases between regions.
- Use synchronous or asynchronous replication based on business requirements.

Failover Mechanism

- If one region fails, traffic automatically switches to another region.
- Users experience little or no downtime.

Monitoring and Alerting

- Monitor CPU, memory, storage, and network.
- Configure automatic alerts for failures.
- Use health checks to detect unhealthy servers.
- Auto-scaling increases capacity during traffic spikes.

Benefits

- High availability
- Fault tolerance
- Faster disaster recovery
- Improved user experience

Conclusion

Multi-region deployment with replication, automatic failover, load balancing, monitoring, and auto-scaling ensures continuous service and protects applications from regional failures.